

POLITECNICO
MILANO 1863

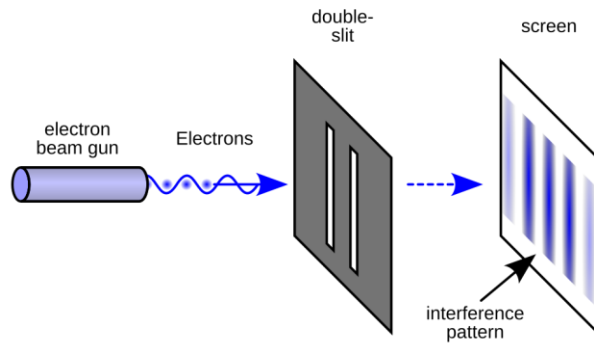
Some Realist and Antirealist positions on the nature of the wave function

Aronne Moltrasio

Course: Philosophy of Science and Technologies
a.y. 2025/2026

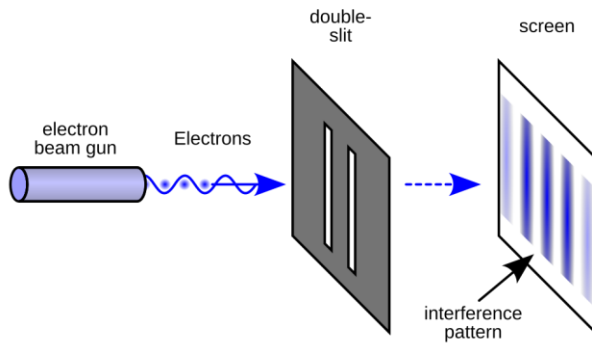
Introduction to Quantum Theory

Particles **are** waves



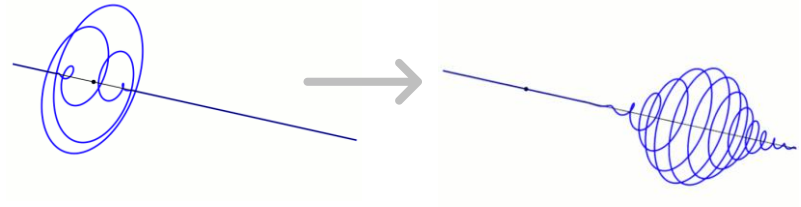
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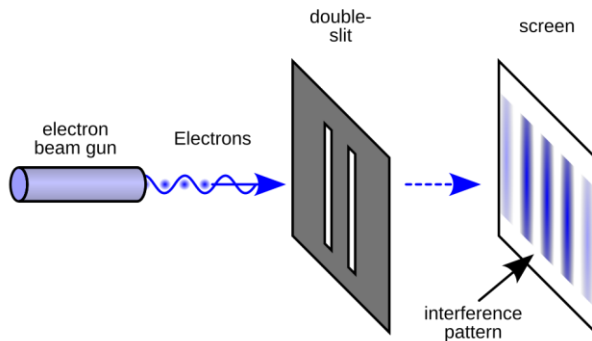
Deterministic time evolution

$$i\hbar \frac{\partial}{\partial t} \Psi(x, t) = \left[-\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} + V(x, t) \right] \Psi(x, t).$$



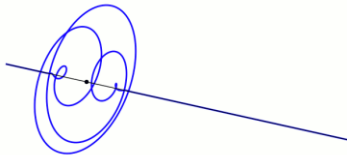
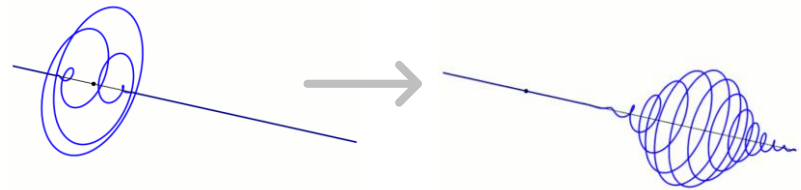
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Measurement



Collapse of ψ

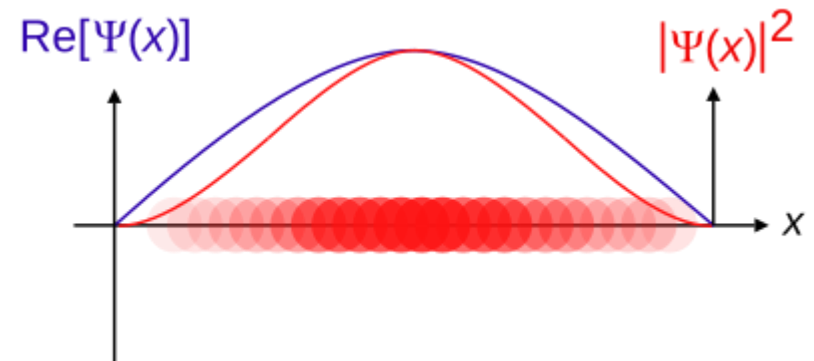
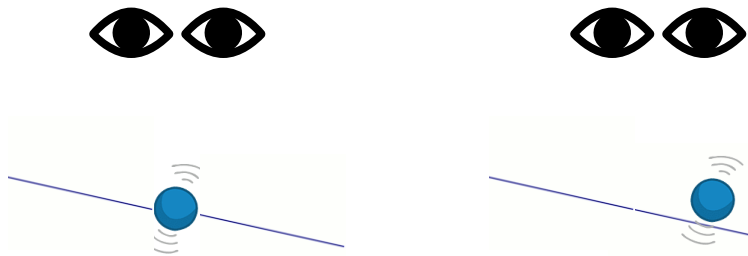


Introduction to Quantum Theory

Objects move and evolve as if they were waves (Ψ wave function).
When observed, waves collapse into a particle state (i.e. position and velocity).
Over many repeated measurements, the collapse is random but follows a probability distribution given by $|\Psi|^2$.

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Questions arising:

- What is an observation?
- What is the dynamic of the collapse?
- What is ψ ?
- ...many others...

ψ is not observable $\rightarrow$ interpretations

Realists believe in an *ontological* nature.

Antirealists believe in an *information bringing* nature.

Bell's inequality violated if one adopts *locality* and *hidden variables*.

⇒ either we abandon locality or hidden variables realism

Ψ predictive success.

Objections: no collapse dynamics

→ Pilot wave (Bohm), Objective collapse, Many worlds

PBR theorem (2012).

Preparation independence and antirealism leads Quantum Theory to an absurd.

(Antirealists criticize PI)

Instrumentalism, no measurement, less ontological objects.
Objections: entanglement, quantum eraser, dynamic of information.

Instrumentalism, no measurement, less ontological objects.

Objections: entanglement, quantum eraser, dynamic of information.

Qbism, instrumentalism.

A decorative horizontal band at the top of the slide, consisting of a series of thin, vertical white lines of varying heights, creating a textured, barcode-like effect.

My opinion.

References & further readings

- ✓ French, Steven, and Juha Saatsi (eds), **Scientific Realism and the Quantum** (Oxford, 2020; online edn, Oxford Academic, 21 May 2020)
- ✓ Claus Kiefer, **Does the quantum mechanical wave function exists?** (arXiv:1908.04607v1 [quant-ph] 13 Aug 2019)
- ✓ **The Stanford Encyclopedia of Philosophy** (Summer 2024 Edition)
- ✓ Bachelor's Degree course "**Complementi di Meccanica Quantistica**" by Franco Ciccacci
- ✓ https://en.wikipedia.org/wiki/Interpretations_of_quantum_mechanics

Thank you!